Seat NO.______ Major______ School______ Student ID □□□□□□□□□□□□ Chinese Name______

（DONOT WRITE YOUR ANSWER IN THIS AREA）

WARNING: MISBEHAVIOR AT EXAM TIME WILL LEAD TO SERIOUS CONSEQUENCE.

SCUT Final Exam

2020-2021-2《Calculus II》 Exam Paper A

Notice:
1. **Make sure that you have filled the form on the left side of seal line.**
2. **Write your answers on the exam paper.**
3. **This is a close-book exam.**
4. **The exam with full score of 100 points lasts 120 minutes.**

Question No.	1-6	7-15	16-21	Sum
Score				

Answer the questions. ($3'\times 6=18'$)

1.The curvature of $y=\ln|\cos x|$ at $x=\frac{\pi}{3}$ is____________. $\frac{1}{2}$

2. Find the line equation pass through A(2,5,1) and B(3,6,4)__________. $\frac{x-2}{1}=\frac{y-5}{1}=\frac{z-1}{3}$

3.Let $D: x^2+y^2\le 2x$, write $\iint_D f(x,y)dxdy$ in polar coordinate, then $\iint_D f(x,y)dxdy=$

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta\int_0^{2\cos\theta} f(r\cos\theta, r\sin\theta)rdr$ ________.

4. If $\vec{a}=\{-1,2,2\},\vec{b}=\{2,-1,2\}$, then $(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})=$____________ $\{12,12,-6\}$ ______.

5.The equation of plane through three points $P_1(1,-2,3),P_2(4,1,-2),P_1(-2,-3,0)$ is

$7x-12y-3z=22.$ ______.

6. The divergence of $\vec{A}=e^{xy}\vec{i}+\cos(xy)\vec{j}+xz^2\vec{k}=$______ $ye^{xy}-x\sin(xy)+2xz$ __.

Finish the following questions. (**7-13:** $7'\times 7=49'$; **14-15:** $6'\times 2=12'$)

7. Evaluate the integral $\iint_D (x^2+2y)\ dxdy$ .where D is the region $y=x^2$, $y=\sqrt{x}$.

Solution:

$$I=\int_0^1 dy\int_{y^2}^{\sqrt{y}}(x^2+2y)dx$$

$$=\int_0^1\left(\frac{1}{3}x^3+2xy\right)\Big|_{y^2}^{\sqrt{y}} dy=\frac{27}{70}.$$

8. Find $\int_{\frac{1}{4}}^{\frac{1}{2}} dx \int_{\frac{1}{2}}^{\sqrt{x}} e^{\frac{x}{y}} dy + \int_{\frac{1}{2}}^{1} dx \int_{x}^{\sqrt{x}} e^{\frac{x}{y}} dy$.

Solution:

$$\mathrm{I} = \iint_D e^{\frac{y}{x}} dxdy = \int_{\frac{1}{2}}^{1} dy \int_{y^2}^{y} e^{\frac{x}{y}} dx = \frac{3}{8} e - \frac{1}{2} e^{\frac{1}{2}}$$

9. Find $\iiint_\Omega (x^2 + y^2 + z^2) dv$, Ω is bounded by $x^2 + y^2 + z^2 = 1$.

Solution:

$$\mathrm{I} = \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_0^1 r^2 \cdot r^2 \sin\varphi dr = \frac{4\pi}{5}$$

10. Find $\iiint_\Omega z dv$, and Ω is bounded by $x^2 + y^2 = 1$ and $z = 0$, $z = 1$.

Solution:

Method A: Cylindrical, $\mathrm{I} = \int_0^{2\pi} d\theta \int_0^1 r dr \int_0^1 z dz = \frac{\pi}{2}$

Method B: Cutting plane method: $\mathrm{I} = \int_0^1 \pi z dz = \frac{\pi}{2}$

11. Let $f(u,v)$ is differentiable, $z=z(x,y)$is determined by $(x+1)z-y^2=x^2 f(x-z,y)$,

find $dz|_{(0,1)}$.

Solution:

Ans:

$x=0,\ y=1 \Rightarrow z=1$

Partial derivatives of x and y on both sides of the equation

$$z+(x+1)\frac{\partial z}{\partial x}=2xf(x-z,y)+x^2 f_1'\cdot(1-\frac{\partial z}{\partial x})$$

$$(x+1)\frac{\partial z}{\partial y}-2y=x^2[f_1'\cdot(-\frac{\partial z}{\partial y})+f_2']$$

when $x=0,\ y=1,\ z=1$

$$\Rightarrow \frac{\partial z}{\partial x}\Big|_{(0,1)}=-1,\ \frac{\partial z}{\partial y}\Big|_{(0,1)}=2$$

$$dz|_{(0,1)}=-dx+2dy$$

12. Find the tangent plane to $z=x^2+y^2$ at $(1,1,2)$.

Solution:

Let $F=x^2+y^2-z$, then $\vec{n}=\langle 2x,2y,-1\rangle$, and $\vec{n}|_{(1,1,2)}=\langle 2,2,-1\rangle$,

so the equation of the tangent plane is

$2(x-1)+2(y-1)-(z-2)=0$

or

$2x+2y-z-2=0.$

13. Find the minimum distance between the original point and the surface $z^2 = x^2 y + 4$.

Solution:

$$z^2 = x^2 y + 4.$$

$$d^2 = x^2 + y^2 + z^2 = x^2 + y^2 + x^2 y + 4,$$

$$f_x = 0,\ f_y = 0 \Rightarrow 2x + 2xy = 0,\ 2y + x^2 = 0,$$

$$\Rightarrow 2x - x^3 = 0 \Rightarrow x = 0 \text{ or } x = \pm\sqrt{2} \Rightarrow y = 0,\ y = -1.$$

$$\text{critical points}: (0,0)\left(\sqrt{2},-1\right)\left(-\sqrt{2},-1\right).$$

$$D(x,y) = f_{xx} f_{yy} - f_{xy}^{\ 2} = (2+2y)2 - 4x^2.$$

$$D(\pm\sqrt{2},-1) = -8 < 0, \text{neither } \left(\sqrt{2},-1\right) \text{nor} \left(-\sqrt{2},-1\right) \text{yiled extremum.}$$

$$D(0,0) = 4 > 0,\ f_{xx}(0,0) = 2 > 0.$$

$$\text{So,}(0,0) \text{ yields the minmum distance . } d^2 = 4. d_{\min} = 2.$$

14. Let $z = f(u,x,y), u = xe^y, f$ has the second-order continuous partial derivative, find $\dfrac{\partial^2 z}{\partial x \partial y}$.

Solution:

$$\frac{\partial z}{\partial x} = f_1' \cdot \frac{\partial u}{\partial x} + f_2' = f_1' \cdot e^y + f_2'$$

$$\Rightarrow \frac{\partial^2 z}{\partial x \partial y} = (f_{11}'' \cdot \frac{\partial u}{\partial y} + f_{13}'')e^y + f_1' \cdot e^y + f_{21}'' \cdot \frac{\partial u}{\partial y} + f_{23}''$$

$$= f_{11}'' \cdot xe^{2y} + f_{13}'' \cdot e^y + f_1' \cdot e^y + f_{21}'' \cdot xe^y + f_{23}''$$

$$= e^y f_1' + xe^{2y} f_{11}'' + e^y f_{13}'' + xe^y f_{21}'' \cdot + f_{23}''.$$

15. Evaluate the integral $\int_0^1 \int_0^{\sqrt{1-y^2}} \sin(x^2+y^2)\mathrm{d}x\mathrm{d}y$.

Solution:

$$I=\int_0^{\frac{\pi}{2}} d\theta \int_0^1 (\sin r^2) r dr$$
$$=\frac{\pi}{4}\int_0^1 \sin r^2 d(r^2)$$
$$=\frac{\pi}{4}(1-\cos 1).$$

Please select 3 questions from the following: $(7'\times 3=21')$

16. Please show that $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{\sqrt{n}}$ is conditionally convergent.

Solution:

By the Alternating Series Test,

$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{\sqrt{n}}$ converges.

Since $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$ is a p-series with $p=\frac{1}{2}$.

$\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$ diverges. So, it is conditionally convergent.

17. Expand the function $f(x)=\arctan\frac{1+x}{1-x}$ into power series of x.

Solution：

$$f(x)=\arctan\frac{1+x}{1-x}\Rightarrow f(0)=\arctan 1=\frac{\pi}{4}$$

$$f'(x)=\frac{1}{1+\left(\frac{1+x}{1-x}\right)^2}\cdot\frac{1\cdot(1-x)-(1+x)\cdot(-1)}{(1-x)^2}$$

$$=\frac{1}{1+x^2}(-1<x<1)$$

$$\text{by}\int_0^x f'(t)\,dt=f(t)\Big|_0^x=f(x)-f(0)=f(x)-\frac{\pi}{4}.$$

$$\Rightarrow f(x)=\frac{\pi}{4}+\int_0^x f'(t)\,dt$$

$$=\frac{\pi}{4}+\int_0^x\frac{1}{1+t^2}\,dt$$

$$=\frac{\pi}{4}+\int_0^x\sum_{n=0}^{\infty}(-1)^n t^{2n}\,dt$$

$$=\frac{\pi}{4}+\sum_{n=0}^{\infty}\int_0^x(-1)^n t^{2n}\,dt$$

$$=\frac{\pi}{4}+\sum_{n=0}^{\infty}(-1)^n\frac{1}{2n+1}x^{2n+1}.(-1\le x<1)$$

18. Find convergent region and sum function of power series $\sum_{n=0}^{\infty}\frac{x^{2n+2}}{(n+1)(2n+1)}$.

Solution：

Ans :

$$\because \lim_{n\to\infty}\frac{\left|\frac{x^{2n+4}}{(n+2)(2n+3)}\right|}{\left|\frac{x^{2n+2}}{(n+1)(2n+1)}\right|}=x^2$$

$\Rightarrow$ when $|x|<1$, absolute convergence; when $|x|>1$, divergence

when $x=\pm1$, series $\sum_{n=0}^{\infty}\frac{1}{(n+1)(2n+1)}$ is convergence.

$\Rightarrow$ Convergence region is $[-1,1]$.

let $f(x)=\sum_{n=0}^{\infty}\frac{x^{2n+2}}{(n+1)(2n+1)}, x\in[-1,1]$

$$f'(x)=2\sum_{n=0}^{\infty}\frac{x^{2n+1}}{2n+1},\ f''(x)=2\sum_{n=0}^{\infty}x^{2n}=\frac{2}{1-x^2}, x\in(-1,1),$$

$\because\ f'(0)=0,\ f(0)=0$

$$\Rightarrow\ f'(x)=\int_0^x f''(t)dt=\int_0^x\frac{2}{1-t^2}dt=,$$

$$f(x)=\int_0^x f'(t)dt=\int_0^x[\ln(1+t)-\ln(1-t)]dt$$

$$=(1+x)\ln(1+x)+(1-x)\ln(1-x), x\in(-1,1)$$

$$f(1)=\lim_{x\to1^-}f(x)=2\ln2,\ f(-1)=\lim_{x\to-1^+}f(x)=2\ln2$$

19. Let L be the circle $x^2+y^2=9$ with counterclockwise direction, find the line integral

$$\oint_L(2xy-2y)\mathrm{d}x+\left(x^2-4x\right)\mathrm{d}y.$$

Solution：

$$I=\iint_D\frac{\partial\left(x^2-4x\right)}{\partial x}-\frac{\partial(2xy-2y)}{\partial y}dA=-2\iint_D dA=-18\pi.$$

20. Find $\iint\limits_{\Sigma} xyz\mathrm{d}S$, here Σ is the boundary of the solid bounded by the coordinate planes and $x+y+z=2$.

Solution:

Let Σ_1: $x=0, y\in R, z\in R$; Σ_2: $y=0, x\in R, z\in R$;

Σ_3: $z=0, x\in R, y\in R$; Σ_4: $x+y+z=1$;

and $\Sigma=\Sigma_1+\Sigma_2+\Sigma_3+\Sigma_4$

so

$$\begin{aligned}\iint_{\Sigma} xyz\mathrm{d}S &= \left(\iint_{\Sigma_1}+\iint_{\Sigma_2}+\iint_{\Sigma_3}+\iint_{\Sigma_4}\right)xyz\mathrm{d}S \\ &= \iint_{\Sigma_4} xyz\mathrm{d}S \\ &= \sqrt{3}\int_0^2 ydy\int_0^{2-y} x(2-x)-xydx \\ &= \sqrt{3}\int_0^2 \frac{1}{6}y(2-y)^3 dy \\ &= \frac{4\sqrt{3}}{15}.\end{aligned}$$

21. Evaluate $I = \iint\limits_{\Sigma} \left[(x^3 z + x)\cos\alpha - x^2 yz\cos\beta - x^2 z^2 \cos\gamma \right] dS$, here Σ is the surface $z = 4 - x^2 - y^2, 1 \le z \le 2$.

Solution:

Assume Σ has outside normal, and Σ_1: $z = 1, n_1$ is downward; Σ_1: $z = 2, n_1$ is upward;

$$
\begin{aligned}
I &= \iint_{\Sigma+\Sigma_1+\Sigma_2} - \iint_{\Sigma_1} - \iint_{\Sigma_2} \\
&= \iiint_{\Omega} 1 dv - \iint_{x^2+y^2\le 3} x^2 dxdy + 4\iint_{x^2+y^2\le 2} x^2 dxdy \\
&= \int_1^2 dz \iint_{D_z} dxdy - \int_0^{2\pi} \cos^2\theta d\theta \int_0^{\sqrt{3}} r^3 dr + 4\int_0^{2\pi} \cos^2\theta d\theta \int_0^{\sqrt{2}} r^3 dr \\
&= \int_1^2 \pi(4-z)dz - \pi \cdot \frac{1}{4} 3^2 + 4\pi \cdot \frac{1}{4} 2^2 \\
&= \frac{5}{2}\pi - \frac{7}{4}\pi \\
&= \frac{3}{4}\pi.
\end{aligned}
$$

If we consider the surface has inside normal, the result should be negative.